

Chapter ① Introduction to Differential Equations

1.1 Definitions and Terminology

A differential equation (DE) is an equation that involves some unknown function and one or more of its derivatives.

Derivatives are taken with respect to some independent variables.

For example ,

$$\frac{dy}{dx} + 3xy = e^x \quad \text{and} \quad y'' + 3y' + 2y = \sin x$$

are differential equations.

In both cases :

$$\begin{cases} y \longrightarrow \text{dependent variable (the unknown)} \\ x \longrightarrow \text{independent variable.} \end{cases}$$

We classify differential equations by:
type, order, and linearity.

(i) Type

- An ordinary differential equation (ODE) is a DE that includes only one independent variable.

ex: $\underbrace{\frac{dx}{dt} + 2x = t^3}_{t \rightarrow \text{indep. variable}}$ and $\underbrace{y'' + y' + 3y = e^x}_{x \rightarrow \text{indep. variable}}$

are ODEs.

Math 252 will be exclusively about
ODEs.

- A partial differential equation (PDE) is a DE that includes more than one independent variables.

ex: $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ and $\frac{\partial u}{\partial y} = \frac{\partial u}{\partial x} + x \cos y$

are PDEs.

This course will not deal with PDEs.

Multivariable Calculus is a prerequisite to study PDEs.

(ii) Order

The order of a DE is the order of the highest derivative in the equation.

ex:

$$\frac{dy}{dx} + 3y = x^3 \rightarrow \text{order} = 1$$

$$\frac{d^2 y}{dx^2} + 2 \left(\frac{dy}{dx} \right)^5 + 3y = \cos x \rightarrow \text{order} = 2$$

$$y^{(5)} + y''' + 2y^6 = x^7 \rightarrow \text{order} = 5$$

Note: y^n means "y power n"

$y^{(n)}$ means "the n^{th} derivative of y with respect to the independent variable"

(iii) Linearity

An n^{th} - order Linear DE is of the

form :

$$a_n(x) y^{(n)} + \dots + a_2(x) y'' + a_1(x) y' + a_0(x) y = f(x)$$

or is equivalent to it.

Note: The coefficients $a_i(x)$ and $f(x)$ are functions of the independent variable x .

ex: (a) $\underbrace{(1+x^3)}_{a_2(x)} y'' + \underbrace{3x}_{a_1(x)} y' + \underbrace{2}_{a_0(x)} y = \underbrace{x^5}_{f(x)}$

It is a Linear DE of order 2.

The DE has the form:

$$a_2(x) y'' + a_1(x) y' + a_0(x) y = f(x)$$

$$(b) \quad \frac{d^2 y}{dx^2} + \underbrace{\left(\frac{dy}{dx} \right)^3}_{\downarrow} + 5y = x^2$$

This makes the DE nonlinear.

It is a nonlinear DE of order 2.

$$(c) \quad y'' + \underbrace{y y'}_{\downarrow} + 2y = 5x$$

This makes the DE nonlinear.

It is a nonlinear DE of order 2.

$$(d) \quad x^2 dy + (y + 2xy - x \sin x) dx = 0$$

is a DE in differential form.

It is equivalent to :

$$x^2 \frac{dy}{dx} + y + 2xy - x \sin x = 0$$

$$\Rightarrow \underbrace{x^2}_{a_1(x)} \frac{dy}{dx} + \underbrace{(1+2x)}_{a_0(x)} y = \underbrace{x \sin x}_{f(x)}$$

It is a Linear DE of order 1.

$$(e) \quad y''' + \underbrace{\sin y}_{\downarrow} = 0$$

this makes the DE nonlinear.

It is a nonlinear DE of order 3.

$$(f) \quad y'' + \sin x = 0$$

It is a Linear DE of order 2.

The DE is equivalent to :

$$y'' + 0y' + 0y = -\sin x.$$

Solutions of DEs

A function is a solution of a DE if, upon substitution of the function and its derivatives into the equation, the equation is satisfied for each value of the independent variable in some interval.

ex: Consider the DE $\frac{dy}{dx} = 2xy$.

The function $y = e^{x^2}$ is a solution of the DE over the interval $(-\infty, \infty)$.

check: $\left\{ \begin{array}{l} \frac{dy}{dx} = 2x e^{x^2} \\ 2xy = 2x e^{x^2} \end{array} \right\} \begin{array}{l} \text{equal for all } x \\ \text{in } (-\infty, \infty). \end{array}$

The DE $\frac{dy}{dx} = 2xy$ has infinitely

many solutions.

$y = ce^{x^2}$ is a solution for any value of c .

$$y = e^{x^2} \quad (c=1)$$

$$y = 7e^{x^2} \quad (c=7)$$

$$y = 0 \quad (c=0)$$

} They are
Particular
Solutions.

$y = ce^{x^2}$ is the General solution

of the DE.

The general solution is a one-parameter family of solutions.

The general solution of an n^{th} order DE includes n arbitrary constants usually denoted by : $c_1, c_2, \dots, c_n$.

ex: Consider the 2^{nd} order DE

$$y'' + 4y = 0.$$

The general solution is :

$$y = c_1 \cos(2x) + c_2 \sin(2x)$$

Some particular solutions are :

$$y = \cos(2x) \quad [c_1 = 1, c_2 = 0]$$

$$y = 5 \cos(2x) - 8 \sin(2x) \quad [c_1 = 5, c_2 = -8]$$

Explicit vs Implicit Solutions

A solution of a DE expressed as

$$y = f(x)$$

is an explicit solution.

ex: $y = e^{x^2}$ is an explicit solution of

$$\frac{dy}{dx} = 2xy.$$

When we solve a DE, we can't always get an explicit solution. A solution sometimes looks like:

$$y^3 + \cos(xy) + xe^y = 0.$$

We can't solve for y explicitly in terms of x . Such a solution is called an implicit solution.

Implicit differentiation can be used to verify implicit solutions.

ex: Verify that $xy = \ln y + C$

is an implicit solution of the DE

$$\frac{dy}{dx} = \frac{y^2}{1 - xy}.$$

Start from: $xy = \ln y + C$.

Take the derivative with respect to x on both sides and solve for $\frac{dy}{dx}$.

$$\Rightarrow \underbrace{y + x \frac{dy}{dx}}_{\text{Product rule}} = \underbrace{\frac{1}{y} \frac{dy}{dx}}_{\text{chain rule}} + 0$$

$$\Rightarrow y^2 + xy \frac{dy}{dx} = \frac{dy}{dx}$$

$$\Rightarrow y^2 = \frac{dy}{dx} - xy \frac{dy}{dx}$$

$$\Rightarrow y^2 = \frac{dy}{dx} (1 - xy)$$

$$\Rightarrow \boxed{\frac{dy}{dx} = \frac{y^2}{1 - xy}}$$

Then, $xy = \ln y + C$ is a

solution of $\frac{dy}{dx} = \frac{y^2}{1 - xy}$.